Project Report Template

You will need to use the IEEE template (A4 Microsoft Word) available for download at following link: https://www.ieee.org/conferences/publishing/templates.html

The IEEE template constitutes a research paper style, which you need to follow. For your convenience following are some information/requirements about the main sections in your report:

- **Abstract**
  The abstract is a brief summary of your work. It should be written to make the reader want to read the rest of your report. Briefly state the basic contents and conclusions of your report: the problem you are solving, why the reader should care about this problem, your unique solution and/or implementation, and the main results and contributions of your work.
- **Introduction**
  The introduction is the big picture of your work: what, why, and how. It includes a definition of the problem you are solving, a high-level description of your solution including any novel techniques and results you provide, and a summary of the main results of your work. In addition, motivates the problem you are solving (why should a reader find your work important), and describes your contribution to the area (this may or may not be applicable to your project).

  - The first paragraph of the introduction should contain all of this information in a very high-level. Subsequent paragraphs should discuss in more detail the problem you are solving, your solution, and your results and conclusions.
  - Statement of Problem Being Solved
  - Motivation
  - Problem Solution
  - Results and Conclusions
- **Related Work**
  This is an essential part of your report; discussing related work is a good way to put your work in context with other similar work, and to provide a way for you to compare/ contrast your work to other's work whenever applicable.
- **One or more sections describing your Solution (e.g. Methodology)**
  - Details of the problem you are solving
  - Details of your solution and the project's implementation
    Even though you may have spent an enormous amount of time writing code, this should not include a listing of any code you wrote. Only if your project is about developing an algorithm or a new language, then may code examples be appropriate here.
  - Discussion of how your solution solves the problem.

- **Experimental Results demonstrating/proving your solution**
  - Explain the tests you performed (and why)
  - Explain how you gathered the data and details of how your experiments were run (any system/environment setup)
  - Present your results
    Choose quality over quantity; the reader will not be impressed with pages and pages of graphs and tables, instead he/she wants to be convinced that your results show something interesting and that your experiments support your conclusions.
  - Discuss your results!
    Explain/interpret your results (possibly compare your results to related work). Do not just present data and leave it up to the reader to infer what the data show and why they are interesting.
- **Conclusions & Future Directions:** for your work, conclude with the main ideas and results of your work. Discuss ways in which your project could be extended...what's next? what are the interesting problems and questions that resulted from your work?
- A brief meta-discussion of your project: Include two paragraphs in this section:
  - Discussion of what you found to be the most difficult and least difficult parts of your project.
  - In what ways did your implementation vary from your proposal and why?
- **References**
  At the end of your report is a Reference section. You must cite each paper/website/textbooks/etc. that you have referenced...your work is related to some prior work.

Writing Style Guidelines

- Write in a top-down style
  First present the high-level issues, then expand them. This applies to the overall organization of your report as well as the organization of sub-sections and individual paragraphs.
- Conclude each paragraph, section and entire report
- Use active 3rd person: example: Example: We present, we show, we demonstrate...
- Define terms, and always define them before using them.
- Use figures/tables with appropriate and descriptive captions:
  Use diagrams to help explain system design, and graphs or tables for presenting results. If your project has a GUI component, then your report should have some screen dumps of your interface. Also, you should have a figure showing the high-level design of your implementation.

Detailed Report and Results Grading Rubric [17.5 points]

Aspects	Excellent 5 points	Good 4 points	Poor 3 points	Score	
1. Strictly follow the provided template				[0-5]	20%
2. Background and research question were clearly stated in the introduction and the logic and argument were reasonable				[0-5]	
3. Contribution of the work was properly articulated in the introduction				[0-5]	
4. Sufficient discussion of state-of-the-art in related work section				[0-5]	
5. Clear, comprehensive, and consistent description of the proposed method (i.e. methodology) and experimental design and experimental dataset				[0-5] x 3	60%
6. Precise execution of quantitative performance evaluation and accuracy metrics				[0-5] x 3	
7. Thorough latency and execution speed analysis				[0-5] x 3	
8. Robust handling, testing, and analysis of failure cases				[0-5] x 3	
9. Detailed analysis, discussion and interpretation of overall experimental results				[0-5] x2	20%
10. Summarization of the work, reasonable discussion of limitation of the proposed solution and future work				[0-5] x2	
			Total	100	
			Total %	100%	

Submitted Code Rubric [7 points]

Important Note: *using someone else code and adopted for your problem is OK but you are not allowed to take someone else code without major modification.*

Aspects	Excellent 5 points	Good 4 points	Poor 3 points	Score
1. Readability	Extensive documentation with Info about programmer, overall program function and interim comments to describe the steps in each function module	Some documentation Info about programmers, program function	No Info about programmer, program function provided	[0 -5] x 2
2. Modularity	The program is completely modular more than one level of function calls	The program is somewhat modular with several function calls	The whole program consists of the main module only.	[0 -5]
3. Efficiency	The program uses structured programming effectively and fits a reasonable length of code	The program uses reasonable length of code	The program is very inefficient and lengthy.	[0 -5] x 2
4. Usability/ Correctness	The program fulfils the requirement of the project with inclusion of at least one from-scratch method and is correct	The program adheres to part of the project, with partial inclusion of at least one from-scratch method, and is partially correct	The program does not solve the original problem or is incorrect and no inclusion of at least one from-scratch method	[0 -5] x 2
5. Robustness	The program shows its ability to handle any possibility of users' erratic interaction	The program shows limited ability to handle some of users' erratic interaction	The program does not handle users' erratic interaction	[0 -5]
6. **GUI** User Interface	GUI Interface is neat and well organized	GUI Interface is reasonably neat	No GUI, and Interface is untidy and not organized	[0 -5] x 2
7. Run without error (50%)	Runs smoothly and without glitches	Runs but with some glitches	Still contain error or warning.	**[0 - 5] x 10**
			Total	100
			Total %	100%

Presentation Preparation Guidelines

Please plan to prepare/deliver a 10-15 minutes' presentation and allow about 5 minutes for questions.

1. Understand Your Audience:

- **Who are they?** What do they know about the topic? What are their interests and needs?
- **Tailor your content** to their level of understanding and expectations.

2. Structure Your Presentation:

- **Create a clear outline.** This will help you organize your thoughts and ensure a logical flow.
- **Start strong.** Grab your audience's attention with a compelling introduction.
- **End memorably.** Leave a lasting impression with a strong conclusion.

3. Practice Makes Perfect:

- **Rehearse multiple times.** This will help you become familiar with your material and identify areas for improvement.
- **Time yourself.** Ensure your presentation fits within the allotted time.
- **Practice in front of others.** Get feedback from friends or family to identify areas for improvement.

4. Visual Aids:

- **Use slides or props** to enhance your presentation.
- **Keep it simple.** Avoid overcrowding slides with too much text or information.
- **Use visuals effectively** to support your points and make your presentation more engaging.

5. Delivery:

- **Speak clearly and confidently.** Project your voice and maintain eye contact with your audience.
- **Use gestures and body language** to emphasize your points.
- **Be enthusiastic.** Your passion for the topic will be contagious.

6. Handle Questions Gracefully:

- **Anticipate potential questions.** Prepare answers in advance.
- **Listen carefully** to the question before responding.
- **Answer concisely and clearly.** Avoid rambling or going off on tangents.

7. Be Prepared for Technical Difficulties:

- **Have a backup plan** in case of technical issues.
- **Test your equipment** beforehand to ensure it is working properly.

Remember, the goal of a presentation is to inform, persuade, or entertain your audience. By following these guidelines, you can deliver a confident, engaging, and effective presentation.

Project Presentation Rubric [10.5 points]

Aspects	Excellent (5)	Good (4)	Poor (3)	Score
Technical Information clearly presented; Concise and Easy to follow	Produces very creative slides that are well-laid out with minimal clutter, no misspellings or grammatical errors, makes use of well formatted graphs, tables, and diagrams to explain and interpret information	Produces a generally good set of slides that conveys the main points well with few misspellings and grammatical errors, makes use of graphs, tables, and diagrams.	Produces slides that convey only the main points, with some misspellings and grammatical errors, makes some use of graphs, tables and diagrams Produces slides that are dull with no creativity and with misspellings and grammatical errors, makes occasional use of graphs, tables, and diagrams Produces slides that are poor, illogically sequenced with numerous mistakes	[0-5] x 5
Deliver a presentation	Speaks in a clear voice with confidence, maintains eye contact, seldom referring to notes, and makes excellent use of volume, pace and gestures.	Speaks in a clear voice, well-paced although monotonic, maintains eye contact most of the time, occasionally referring to notes, and makes some use of gestures	Speaks in a low voice, often refers to notes, occasionally uses eye contact and gestures Delivers in a low, sometimes inaudible voice, always refers to notes, makes minimal eye contact and gestures Delivers in an inaudible voice, basically reading from notes, with too many distracting gestures and no eye contact	[0-5] x 5
Discussion: and Team Work	Actively participated in question-answering sessions and group discussions, demonstrating a deep understanding of the topic. Contributed meaningfully to the team's overall performance.	Participated in discussions and answered questions appropriately. Showed a solid understanding of the topic but could have contributed more to the group's dynamic.	Participated in discussions but contributions were limited. Demonstrated a basic understanding of the topic. Participated minimally in discussions and struggled to answer questions. Understanding of the topic was unclear. Did not participate in discussions or answer questions. Showed little to no understanding of the topic.	[0-5] x 5
Demonstration of design functionalities and individual ability to explain, reproduce, and live-modify code during the oral presentation.	Clearly and effectively demonstrated all design functionalities, highlighting their strengths and benefits. Presentation was well-organized and easy to follow.	Demonstrated most design functionalities, but could have provided more detail or clarity in certain areas. Presentation was well-structured, but could have been more engaging.	Demonstrated some design functionalities, but presentation was unclear or incomplete. Could have provided more evidence of the design's effectiveness. Demonstrated limited design functionalities, and presentation was disorganized or difficult to follow. Could not clearly explain the benefits of the design. Failed to demonstrate any design functionalities or provide a clear explanation of the design's purpose.	[0-5] x 5
			Total	100
			Total %	100%